



Launching VPC Resources



MHM Aslam

The screenshot displays the AWS Management Console 'Create VPC' wizard, specifically step 2: 'Customize subnets CIDR blocks'. The interface is divided into two main sections: configuration options on the left and a network topology preview on the right.

Configuration Options (Left):

- NAT gateways (5) - updated**: A notification indicates that NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply. The 'Regional - new' option is selected.
- Introducing regional NAT gateway**: A blue box highlights that AWS now offers a multi-AZ NAT Gateway, eliminating the need for separate NAT Gateways across availability zones.
- VPC endpoints**: A notification states that endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. The 'S3 Gateway' option is selected.
- DNS options**: Both 'Enable DNS hostnames' and 'Enable DNS resolution' are checked.
- Additional tags**: A section for adding tags is visible at the bottom.

Preview (Right):

The preview pane shows the network topology for the VPC. It includes:

- Subnets (4)**: Subnets within this VPC.
 - us-east-1a**:
 - nextwork-subnet-public1-us-east-
 - nextwork-subnet-private1-us-east-
 - us-east-1b**:
 - nextwork-subnet-public2-us-east-
 - nextwork-subnet-private2-us-east-
- Route tables (3)**: Route network traffic to resources.
 - nextwork-rtb-public
 - nextwork-rtb-private1-us-east-1a
 - nextwork-rtb-private2-us-east-1b

A circular profile picture of a person with short dark hair, wearing a blue and white sports jersey, standing in front of a green background with some text and graphics.

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is your own private network inside AWS cloud. You can pick IP addresses, create public subnets for web servers, and private subnets for databases. It's useful because you control who goes in and out using internet gateways, security groups, and NACLs. Your resources stay safe, organized, and hidden from the internet if you want. Without VPC, cloud setup is messy. With VPC, you get full control. Simple as that!

How I used Amazon VPC in this project

I used Amazon VPC to build a complete cloud network from scratch — first by creating a custom VPC with public and private subnets, an internet gateway, route tables, security groups, and network ACLs. Then I launched one EC2 instance inside the public subnet (so it can be reached from the internet) and another EC2 instance inside the private subnet (so it stays hidden and safe). I also set up security rules so only the public instance can talk to the private one using SSH. Later in the project, I used the VPC wizard to create a second VPC super fast — with just a few clicks, it made subnets, route tables, and gateways automatically. I learned how to choose CIDR blocks, how NAT gateways work, and why AWS limits public subnets to 0 or 2 for high availability. So basically, VPC was the foundation for everything I did today.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was that you can have two VPCs with the exact same CIDR block (like 10.0.0.0/16) in the same AWS account and region without any problem. I thought IP addresses would clash like in a home network, but because VPCs are totally isolated unless you connect them, it's actually fine. That surprised me!



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This project took me...

This project took me about 1.5 to 2 hours — including reading the steps, launching both EC2 instances, and trying out the VPC wizard. But if I do it again, I could finish much faster because now I understand the flow.



Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to it from your own computer over the internet — like using Remote Desktop or SSH to get inside that server and run commands, install software, or check files. It's like sitting in front of that computer, but actually it's running somewhere in AWS cloud.

SSH is a key method for directly accessing a VM

SSH stands for Secure Shell. It's a protocol that lets you securely connect from your computer to a remote machine (like an EC2 instance) over the internet. All the data you send and receive is encrypted, so no one can spy on it. SSH traffic means network communication that uses the SSH protocol — like when you run commands on a server, SSH traffic carries those commands safely between your computer and the server.

To enable direct access, I set up key pairs

A key pair is like a lock and key for your EC2 instance. It has two parts: a public key (stays on AWS) and a private key (you download and keep). When you connect to your instance, AWS checks if your private key matches the public key — if yes, you get in. It's a secure way to log in without using a password.

A private key's file format means the way the private key is saved and structured inside a file — like how you save text as .txt or image as .jpg. Different formats work with different software. My private key's file format was .pem (Privacy Enhanced Mail), which is widely supported by AWS and many servers



Launching a public server

I had to change my EC2 instance's networking settings by clicking Edit in the Network settings panel, then selecting NextWork VPC from the VPC dropdown, choosing my public subnet, and for the firewall I selected "Select existing security group" and picked NextWork Public Security Group.

Instances (1/1) [Info](#)

Last updated less than a minute ago

[Connect](#)

[Instance state](#)

[Actions](#)

[Launch instances](#)

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	NextWork Public Server	i-09b87883f6bd9f90e	Running	t2.micro	Initializing	us-east-1e	-

i-09b87883f6bd9f90e (NextWork Public Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID

vpc-02efa2fbcc0d2ecfc (NextWork VPC)

Subnet ID

subnet-0c77b543c6d069c63 (NextWork Public Subnet)

Availability zone

us-east-1e

Availability zone ID

use1-az3

Outpost ID

-

IP addresses

Info

Public IPv4 address

100.27.2.141 | [open address](#)

Private IPv4 addresses

10.0.0.226

IPv6 addresses

-

Secondary private IPv4 addresses

-

Carrier IP addresses (ephemeral)

-



Launching a private server

My private server has its own dedicated security group because it should not be directly accessible from the internet. The public server needs to allow HTTP and SSH from anywhere (0.0.0.0/0), but the private server only allows traffic from the NextWork Public Security Group — meaning only my public EC2 instance can talk to it. This keeps the private server safe and hidden.

My private server's security group's source is NextWork Public Security Group, which means only resources that belong to that security group (like my public EC2 instance) are allowed to talk to my private server — no one from the open internet can reach it.

Security group name - *required*

NextWork Private Security Group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .-:/()#,@[]+=&:{}!\$*

Description - *required*

Info

Security group for NextWork Private Subnet.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-0ef76248cf3666aca)

Type

Info

ssh

Protocol

Info

TCP

Port range

Info

22

Source type

Info

Custom

Source

Info

Q Add CIDR, prefix list or security group

sg-0ef76248cf3666aca

X

Description - *optional*

Info

e.g. SSH for admin desktop

Add security group rule

► Advanced network configuration

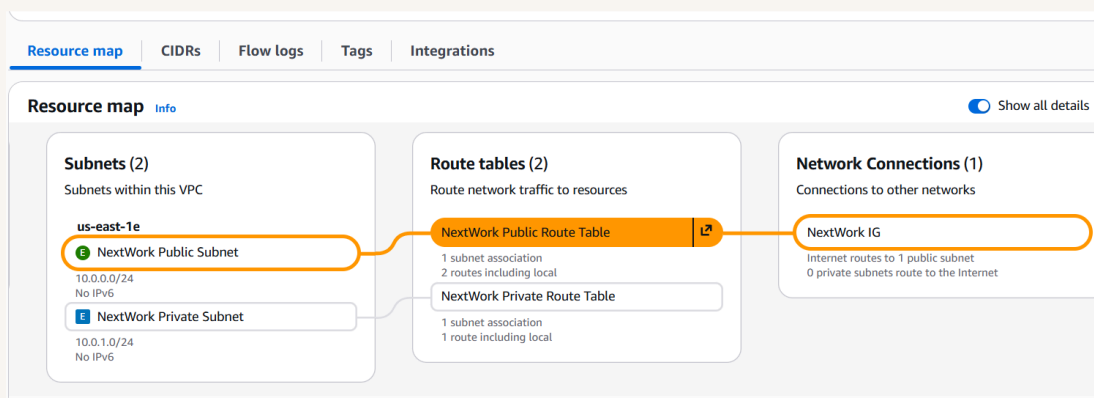


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used the VPC wizard instead of creating everything manually. I turned on name tag auto-generation (so all resources got "nextwork" in their names), set the IPv4 CIDR to 10.0.0.0/16, chose 1 Availability Zone, created 1 public subnet (10.0.0.0/24) and 1 private subnet (10.0.1.0/24), selected No NAT gateways to avoid costs, and kept DNS options enabled. Then I just clicked Create VPC — and in a few minutes, everything was ready!

A VPC resource map is a visual diagram inside the AWS console that shows all the parts of your VPC — like subnets, route tables, internet gateways, and NAT gateways — and how they are connected to each other. It helps you understand your network setup at a glance without clicking through lots of pages.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because VPCs are isolated from each other by default — they don't try to talk to each other unless I connect them using VPC peering. So there's no IP address conflict as long as I keep them separate.



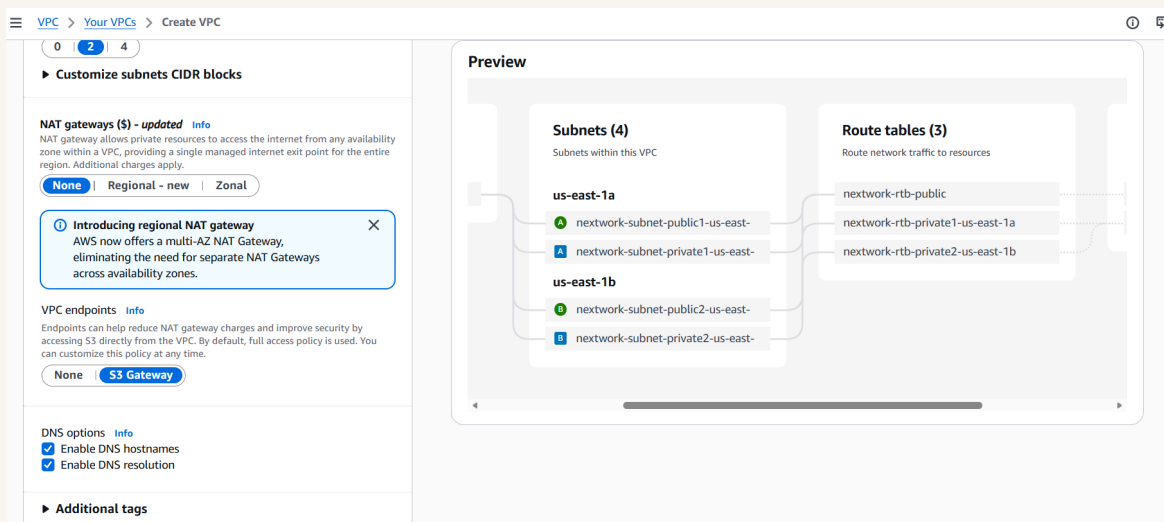


Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: 0 or 2. This was because AWS follows best practices for high availability — if you choose 2 Availability Zones, the wizard forces you to have a public subnet in each zone so your resources stay online even if one zone fails. You can't create just one public subnet because that wouldn't give you redundancy.

The set up page also offered to create NAT gateways, which are devices that let EC2 instances in private subnets access the internet (for things like downloading updates or patches) but block anyone from the internet from reaching those instances. Unlike an internet gateway, a NAT gateway keeps your private instances hidden because they don't need public IP addresses — the NAT gateway does the translation for them. And yes, they cost money (that's why we selected "None" for this project).





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